

```
!pip install transformers peft datasets accelerate bitsandbytes faiss-cpu sentence-transformers
```

```
Requirement already satisfied: packaging in /opt/anaconda3/lib/python3.13/site-packages (from transformers) (24.1)
Requirement already satisfied: pyyaml>=5.1 in /opt/anaconda3/lib/python3.13/site-packages (from transformers) (6.0.1)
Requirement already satisfied: regex!=2019.12.17 in /opt/anaconda3/lib/python3.13/site-packages (from transformers) (2024.9.11)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /opt/anaconda3/lib/python3.13/site-packages (from transformers) (0.20.1)
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Requirement already satisfied: tqdm>=4.27 in /opt/anaconda3/lib/python3.13/site-packages (from transformers) (4.67.1)
Requirement already satisfied: filelock in /opt/anaconda3/lib/python3.13/site-packages (from huggingface-hub<2.0,>=1.0) (3.16.1)
Requirement already satisfied: fsspec>=2023.5.0 in /opt/anaconda3/lib/python3.13/site-packages (from huggingface-hub<2.0,>=1.0) (2024.10.1)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /opt/anaconda3/lib/python3.13/site-packages (from huggingface-hub<2.0,>=1.0) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /opt/anaconda3/lib/python3.13/site-packages (from huggingface-hub<2.0,>=1.0) (0.27.0)
Requirement already satisfied: shellingham in /opt/anaconda3/lib/python3.13/site-packages (from huggingface-hub<2.0,>=1.0) (1.0.0)
Requirement already satisfied: typing-extensions>=4.1.0 in /opt/anaconda3/lib/python3.13/site-packages (from huggingface-hub<2.0,>=1.0) (4.12.0)
Requirement already satisfied: anyio in /opt/anaconda3/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.0) (4.6.2)
Requirement already satisfied: certifi in /opt/anaconda3/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.0) (2024.12.14)
Requirement already satisfied: httpcore==1.* in /opt/anaconda3/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.0) (1.0.7)
Requirement already satisfied: idna in /opt/anaconda3/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.0) (3.10)
Requirement already satisfied: h11>=0.16 in /opt/anaconda3/lib/python3.13/site-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.0) (0.14.0)
Requirement already satisfied: psutil in /opt/anaconda3/lib/python3.13/site-packages (from peft) (7.0.0)
Requirement already satisfied: torch>=1.13.0 in /opt/anaconda3/lib/python3.13/site-packages (from peft) (2.10.0)
Requirement already satisfied: pyarrow>=21.0.0 in /opt/anaconda3/lib/python3.13/site-packages (from datasets) (21.0.0)
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Requirement already satisfied: pandas in /opt/anaconda3/lib/python3.13/site-packages (from datasets) (2.3.3)
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Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /opt/anaconda3/lib/python3.13/site-packages (from datasets) (4.10.0)
Requirement already satisfied: setuptools in /opt/anaconda3/lib/python3.13/site-packages (from torch>=1.13.0->peft) (75.1.0)
Requirement already satisfied: sympy>=1.13.3 in /opt/anaconda3/lib/python3.13/site-packages (from torch>=1.13.0->peft) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /opt/anaconda3/lib/python3.13/site-packages (from torch>=1.13.0->peft) (3.4.1)
Requirement already satisfied: Jinja2 in /opt/anaconda3/lib/python3.13/site-packages (from torch>=1.13.0->peft) (3.1.4)
Requirement already satisfied: scikit-learn>=0.22.0 in /Users/alsayao/.local/lib/python3.13/site-packages (from sentence-transformers) (1.5.2)
Requirement already satisfied: scipy>=1.0.0 in /opt/anaconda3/lib/python3.13/site-packages (from sentence-transformers) (1.14.1)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /opt/anaconda3/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (2.5.0)
Requirement already satisfied: aiosignal>=1.4.0 in /opt/anaconda3/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /opt/anaconda3/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (25.3.0)
Requirement already satisfied: frozenlist>=1.1.1 in /opt/anaconda3/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /opt/anaconda3/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (6.0.0)
Requirement already satisfied: propcache>=0.2.0 in /opt/anaconda3/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (0.2.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /opt/anaconda3/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (1.18.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /opt/anaconda3/lib/python3.13/site-packages (from requests>=2.32.2) (3.4.0)
Requirement already satisfied: urllib3<3,>=1.21.1 in /opt/anaconda3/lib/python3.13/site-packages (from requests>=2.32.2) (2.3.0)
Requirement already satisfied: joblib>=1.3.0 in /opt/anaconda3/lib/python3.13/site-packages (from scikit-learn>=0.22.0) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.2.0 in /opt/anaconda3/lib/python3.13/site-packages (from scikit-learn>=0.22.0) (3.5.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/anaconda3/lib/python3.13/site-packages (from sympy>=1.13.3) (1.3.0)
Requirement already satisfied: sniffio>=1.1 in /opt/anaconda3/lib/python3.13/site-packages (from anyio->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.0) (1.3.1)
Requirement already satisfied: MarkupSafe>=2.0 in /opt/anaconda3/lib/python3.13/site-packages (from Jinja2->torch>=1.13.0->peft) (3.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/anaconda3/lib/python3.13/site-packages (from pandas->datasets) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /opt/anaconda3/lib/python3.13/site-packages (from pandas->datasets) (2024.2)
Requirement already satisfied: tzdata>=2022.7 in /opt/anaconda3/lib/python3.13/site-packages (from pandas->datasets) (2025.1)
Requirement already satisfied: six>=1.5 in /opt/anaconda3/lib/python3.13/site-packages (from python-dateutil>=2.8.2) (1.17.0)
Requirement already satisfied: click>=8.0.0 in /opt/anaconda3/lib/python3.13/site-packages (from typer-slim->transformers) (8.1.8)
Downloading bitsandbytes-0.49.2-py3-none-macosx_14_0_arm64.whl (131 kB)
Using cached faiss_cpu-1.13.2-cp310-abi3-macosx_14_0_arm64.whl (3.5 MB)
Using cached sentence_transformers-5.4.1-py3-none-any.whl (571 kB)
Installing collected packages: faiss-cpu, bitsandbytes, sentence-transformers
Successfully installed bitsandbytes-0.49.2 faiss-cpu-1.13.2 sentence-transformers-5.4.1
```

GPU check

```
import torch
import matplotlib.pyplot as plt
from transformers import (
    AutoTokenizer, AutoModelForCausalLM,
    TrainingArguments, Trainer, TrainerCallback,
    DataCollatorForLanguageModeling, EarlyStoppingCallback
)
from peft import LoraConfig, get_peft_model, TaskType
from datasets import load_dataset

# Confirm GPU
device = "cuda" if torch.cuda.is_available() else "cpu"
print(f"Using device: {device}")
print(f"GPU: {torch.cuda.get_device_name(0) if device == 'cuda' else 'None'}")
```

```
Using device: cuda
GPU: Tesla T4
```

Load model and tokenizer

```
MODEL_NAME = "TinyLlama/TinyLlama-1.1B-Chat-v1.0"

tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)
tokenizer.pad_token = tokenizer.eos_token # required for batching

model = AutoModelForCausalLM.from_pretrained(
    MODEL_NAME,
    torch_dtype=torch.float16, # saves VRAM
    device_map="auto"         # auto-places on GPU
)
print(f"Model loaded. Parameters: {model.num_parameters():,}")
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/token>)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
config.json: 100% 608/608 [00:00<00:00, 58.1kB/s]

tokenizer_config.json: 1.29k/? [00:00<00:00, 95.2kB/s]

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits a
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_
tokenizer.json: 1.84M/? [00:00<00:00, 40.9MB/s]

tokenizer.model: 100% 500k/500k [00:00<00:00, 2.49MB/s]

special_tokens_map.json: 100% 551/551 [00:00<00:00, 63.4kB/s]
`torch_dtype` is deprecated! Use `dtype` instead!
model.safetensors: 100% 2.20G/2.20G [00:19<00:00, 132MB/s]

Loading weights: 100% 201/201 [00:02<00:00, 76.21it/s, Materializing param=model.norm.weight]

generation_config.json: 100% 124/124 [00:00<00:00, 12.8kB/s]

Model loaded. Parameters: 1,100,048,384
```

Apply LoRA

```
lora_config = LoraConfig(
    task_type=TaskType.CAUSAL_LM,
    r=8, # rank - keep low for Colab
    lora_alpha=16, # scaling factor
    lora_dropout=0.1, # regularization (dropout)
    target_modules=["q_proj", "v_proj"], # which layers to adapt
    bias="none"
)

model = get_peft_model(model, lora_config)
model.print_trainable_parameters()
# Expect ~0.1% of params trainable - that's normal and correct
```

trainable params: 1,126,400 || all params: 1,101,174,784 || trainable%: 0.1023

Load & Format Dataset (SFT / Instruction Tuning)

```
# Load Alpaca-style instruction dataset
dataset = load_dataset("yahma/alpaca-cleaned", split="train")

# Filter to explanation/teaching examples for relevance
dataset = dataset.filter(lambda x: len(x["output"]) > 100)
dataset = dataset.shuffle(seed=42).select(range(5000)) # subset for speed

PROMPT_TEMPLATE = """### Instruction:
{instruction}

### Input:
{input}
```

```

### Response:
{output}"""

def format_example(example):
    text = PROMPT_TEMPLATE.format(
        instruction=example["instruction"],
        input=example.get("input", ""),
        output=example["output"]
    )
    tokenized = tokenizer(
        text,
        truncation=True,
        max_length=512,
        padding="max_length"
    )
    tokenized["labels"] = tokenized["input_ids"].copy()
    return tokenized

tokenized_dataset = dataset.map(format_example, remove_columns=dataset.column_names)
split = tokenized_dataset.train_test_split(test_size=0.1, seed=42)
train_dataset = split["train"]
eval_dataset = split["test"]

print(f"Train: {len(train_dataset)} | Eval: {len(eval_dataset)}")

```

README.md: 11.6k/? [00:00<00:00, 942kB/s]
 alpaca_data_cleaned.json: 100% 44.3M/44.3M [00:00<00:00, 221MB/s]
 Generating train split: 100% 51760/51760 [00:00<00:00, 85395.01 examples/s]
 Filter: 100% 51760/51760 [00:00<00:00, 107974.15 examples/s]
 Map: 100% 5000/5000 [00:05<00:00, 1152.51 examples/s]
 Train: 4500 | Eval: 500

Training curve callback

```

# Custom callback to record loss at each step
class LossTracker(TrainerCallback):
    def __init__(self):
        self.train_losses = []
        self.eval_losses = []
        self.steps = []

    def on_log(self, args, state, control, logs=None, **kwargs):
        if logs:
            if "loss" in logs:
                self.train_losses.append(logs["loss"])
                self.steps.append(state.global_step)
            if "eval_loss" in logs:
                self.eval_losses.append(logs["eval_loss"])

loss_tracker = LossTracker()

```

Training arguments and trainer

```

training_args = TrainingArguments(
    output_dir="./models/lora-coach",
    num_train_epochs=3,
    per_device_train_batch_size=4,
    gradient_accumulation_steps=4, # effective batch = 16 (stability)
    per_device_eval_batch_size=4,
    warmup_steps=50,
    eval_strategy="steps",
    eval_steps=100,
    save_strategy="steps",
    save_steps=100,
    logging_steps=20,
    learning_rate=2e-4,
    weight_decay=0.01, # L2 regularization
    fp16=True,
    load_best_model_at_end=True, # for early stopping
    metric_for_best_model="eval_loss",

```

```

        report_to="none"                # disable wandb
    )

    trainer = Trainer(
        model=model,
        args=training_args,
        train_dataset=train_dataset,
        eval_dataset=eval_dataset,
        data_collator=DataCollatorForLanguageModeling(tokenizer, mlm=False),
        callbacks=[
            loss_tracker,
            EarlyStoppingCallback(early_stopping_patience=3) # regularization
        ]
    )

    trainer.train()

```

 [846/846 43:56, Epoch 3/3]

Step	Training Loss	Validation Loss
100	1.070570	1.122959
200	1.098289	1.121467
300	1.080205	1.120184
400	1.118330	1.120701
500	1.075002	1.119334
600	1.072736	1.119198
700	1.047599	1.119100
800	1.101963	1.118789

TrainOutput(global_step=846, training_loss=1.0817494600940822, metrics={'train_runtime': 2639.3845, 'train_samples_per_second': 5.115, 'train_steps_per_second': 0.321, 'total_flos': 4.2950011650048e+16, 'train_loss':

Plot training curves

```

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

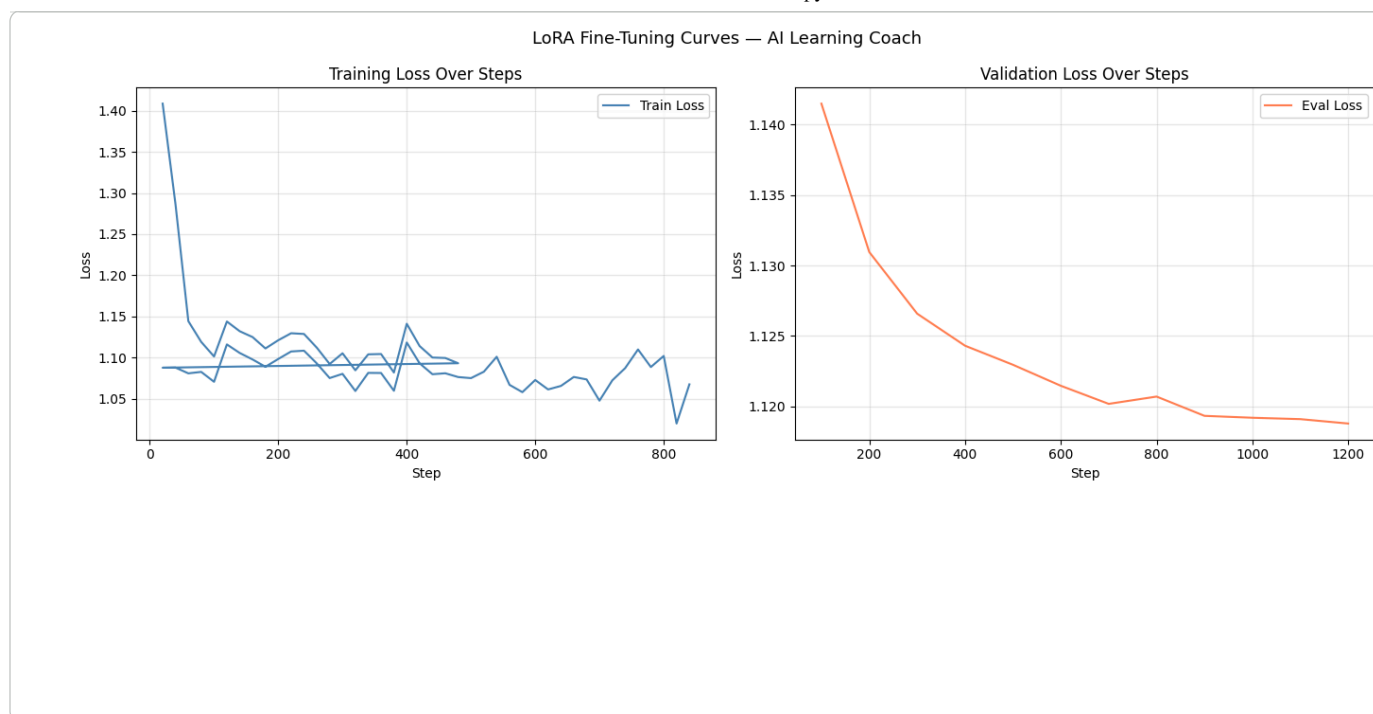
ax1.plot(loss_tracker.steps, loss_tracker.train_losses, color="steelblue", label="Train Loss")
ax1.set_title("Training Loss Over Steps")
ax1.set_xlabel("Step")
ax1.set_ylabel("Loss")
ax1.legend()
ax1.grid(alpha=0.3)

eval_steps_x = [i * 100 for i in range(1, len(loss_tracker.eval_losses) + 1)]
ax2.plot(eval_steps_x, loss_tracker.eval_losses, color="coral", label="Eval Loss")
ax2.set_title("Validation Loss Over Steps")
ax2.set_xlabel("Step")
ax2.set_ylabel("Loss")
ax2.legend()
ax2.grid(alpha=0.3)

plt.suptitle("LoRA Fine-Tuning Curves – AI Learning Coach", fontsize=13)
plt.tight_layout()

plt.show()

```



As we can see, the training loss drastically decreases in the first few training steps 0-200, but from 200-800 steps it fluctuates while still following a general trend of decline in loss. The validation loss, on the other hand, follows a smoother and more consistent decline over steps, showing the improvement of the model and increased accuracy over training.

Save model

```
from google.colab import drive
import os

drive.mount('/content/drive')

save_path = "/content/drive/MyDrive/372-Project/models/lora-coach"

if not os.path.exists(save_path):
    os.makedirs(save_path)

model.save_pretrained(save_path)
tokenizer.save_pretrained(save_path)

print(f"Model successfully saved to Google Drive at: {save_path}")

Model saved.
Mounted at /content/drive
Model successfully saved to Google Drive at: /content/drive/MyDrive/372-Project/models/lora-coach
```

Custom generation with temperature sampling

```
def generate_response(prompt, temperature=0.7, max_new_tokens=256):
    """Custom generation with temperature sampling."""
    inputs = tokenizer(prompt, return_tensors="pt").to(device)

    with torch.no_grad():
        outputs = model.generate(
            *inputs,
            max_new_tokens=max_new_tokens,
            do_sample=True,          # enable sampling
            temperature=temperature, # controls randomness
            top_p=0.9,              # nucleus sampling
            repetition_penalty=1.1,
            pad_token_id=tokenizer.eos_token_id
        )
```

```
decoded = tokenizer.decode(outputs[0], skip_special_tokens=True)
# Strip the prompt from output
return decoded[len(tokenizer.decode(inputs["input_ids"][0], skip_special_tokens=True)):]

# Quick test
test_prompt = """### Instruction:
Explain the concept of gradient descent to a student who is learning for the first time.

### Input:

### Response:"""

print(generate_response(test_prompt))
```

Gradient descent is a popular algorithm used in machine learning and optimization. It works by finding the direction

The output is on topic and grammatically correct, which is good, with display of temperature sampling. However, it's factually imprecise with some incorrect information, such as saying that gradient descent is the direction of least curvature, which is incorrect. This is probably due to the model hallucinating from limited training.

This is our iteration 1 baseline model, and I will implement RAG in the next notebook to improve factual precision and correctness of the outputs to give students accurate information when using this model and reduce hallucinations.

Start coding or [generate](#) with AI.